

UNIT 2

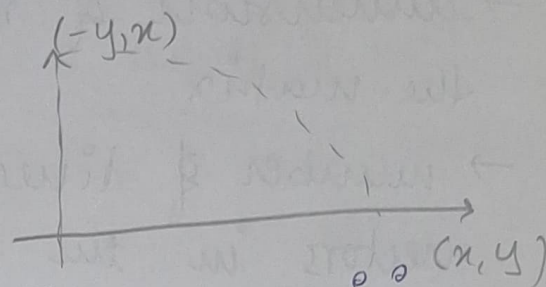
Linear Transformations:

1.

$A = \begin{bmatrix} c & 0 \\ 0 & c \end{bmatrix}$ If $x = (x, y)$ then $Ax = (cx, cy)$
- A multiple of the identity matrix stretches every vector by a factor of c .

2. Rotation:

$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ If $x = (x, y)$ then $Ax = (-y, x)$



The matrix A rotates every vector about origin through a right angle in counter clockwise.

3. Reflection:

$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ - Also a permutation matrix
If $x = (x, y)$ then $Ax = (y, x)$

4. Projection:

$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ If $x = (x, y)$ then $Ax = (x, 0)$

- Matrix projects every vector onto x axis.

Rule of linearity:

$$T(cx + dy) = cT(x) + dT(y)$$

RREF:

- Row reduced echelon form.
- All elements above & below the pivot element should be zero and the pivot element has to be 1.

Rank:

- dimension of row space / column space for the matrix.
- number of linearly independent row / column vectors in the matrix.
- Matrix is invertible only if square & rank is same as number of rows / columns.

Complete Solution:

$$x = x_n + x_p$$

where $x_n \Rightarrow$ solution of $Ax = 0$ [$N(A)$]

$x_p \Rightarrow$ solution of $Ax = b$ [$C(A)$ basis]

Independent Dependent Vectors:

→ Take $c_1x_1 + c_2x_2 + \dots + c_nx_n = 0$ if and only if $c_1, c_2, \dots, c_n = 0$ then linearly independent.

→ If c 's are one non zero then vectors are linearly dependent.

→ For matrix A , $N(A) = \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$ [only zero vector]

→ r rank of matrix A then r columns of A are independent [pivot columns].

Spanning Subspace:

→ If a vector space V consists of all linear combinations of $w_1, \dots, w_r$ then the vectors span the space.

⇒ every vector $v \in V$ is a combo of w 's.

→ $C(A)$ → spanned by columns

$C(A^T)$ → spanned by rows

* spanning = column space *

* independence = null space *

Basis of Vector Space:

- basis are vectors that are [for a space V]

→ linearly independent.

→ span the space of V .

- first columns of matrix A forms basis for $C(A)$

* maximal independent set *

* minimal spanning set *

Dimension of Vector Space:

- number of vectors in any basis of 'V' [the vector space].
- degrees of freedom of the space

Row Space:

- All possible linear combinations of row vectors of a matrix.
- $C(A^T)$, subspace of $\mathbb{R}^n$.
- $\dim(C(A)) = \text{rank}(A)$
- Basis is pivot ~~columns~~^{rows} of row matrix.

Column Space:

- All linear combinations of column vectors of A.
- $C(A)$, subspace of $\mathbb{R}^m$
- $\dim(C(A)) = \text{rank}(A)$
- Basis is pivot columns of matrix.

Null Space:

- All solutions of matrix such that $Ax = 0$.
- $N(A)$, subspace of $\mathbb{R}^n$
- $\dim(N(A)) = n - \text{rank}(A) \Rightarrow \text{NULLITY}$
- Basis of $N(A)$ is solutions to $U(x) = 0$.

Left Null Space:

- All solutions of matrix such that $A^T y = 0$.
- $N(A^T)$, subspace of $\mathbb{R}^m$
- $\dim(N(A^T)) = m - \text{rank}(A)$
- basis of $N(A^T)$ zero rows of U in A .

Rank Nullity Theorem:

For any $A_{m \times n}$,

$$\dim(C(A)) + \dim(N(A)) = \text{no. of columns}$$
$$r + (n - r) = n$$

For $A^T_{n \times m}$,

$$\dim(C(A^T)) + \dim(N(A^T)) = \text{no. of rows}$$
$$r + (m - r) = m$$

Inverses:

- left inverse of $A \Rightarrow BA = I$
- right inverse of $A \Rightarrow AC = I$
- If $\text{rank} = m$, right inverse exists & $Ax = b$ has sol.
- If $\text{rank} = n$, left inverse exists & a unique solution
- * Only square matrices have both one or two sided inverse exists. *

For Rectangular Matrices

Best Left Inverse

$$B = (A^T A)^+ A^T$$

Best Right Inverse

$$C = A^T (A A^T)^+ A$$

Matrix of Rank 1:

- When rank is as small as possible, complex matrix can be written as product of a row $\times$ column.
- These matrices have rank 1.

$$A = uv^T = \text{column} \times \text{row}$$

Linear Transformation:

- Consider $T: A \rightarrow B$ where A, B are subspaces
 - $\rightarrow A$ is domain
 - $\rightarrow B$ is codomain
- $A(cx) = cA(x)$ [Homogeneity]
- $A(x+y) = Ax + Ay$ [Additivity]
- Set of All images, subset of B is range of transformation.

→ For all x in A such that $Tx = 0$, this set is called kernel of transformation.

- $\text{Dim}(\text{range}) = \text{rank}$

$\text{Dim}(\text{kernel}) = \text{nullity}$.

$$* \begin{bmatrix} \text{range} = \text{columnspace} \\ \text{kernel} = \text{nullspace} \end{bmatrix} *$$

Polynomial Space:

- The space of all polynomials in t of degree ' n ' is a vector space denoted by P_n .

- Basis is $1, t, t^2, \dots, t^n$

- Dimension is $n+1$.

Rotation, Reflection, Projection in Trig:

Rotation Matrix:

$$Q_\theta = \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \quad Q_\theta^T = Q_{-\theta} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix}$$

$$Q_\theta^2 = Q_{2\theta}$$

$$Q_\theta \circ Q_\alpha = Q_{(\theta+\alpha)}$$

Projection Matrix :

$$P = \begin{bmatrix} c^2 & cs \\ cs & s^2 \end{bmatrix} \quad P^2 = P$$

Reflection Matrix :

$$H = \begin{bmatrix} 2c^2 - 1 & 2cs \\ 2cs & 2s^2 - 1 \end{bmatrix} \quad \begin{aligned} H &= H^{-1} \\ H^2 &= I \end{aligned}$$

- Two reflections bring back the original.

Relation b/w Reflection & Projection :

$$H = 2P - I$$